

ISYS1055/3412 (Practical) Database Concepts

Assessment 4: Database Design Project



Assessment type: Take-home assessment

Word limit: N/A



Draft Due Date: 20 October at 23:59 (Melbourne Time) – Week 12 (otherwise no 4 mark bonus)

Final Due Date: 3 November at 23:59 (Melbourne Time) – Week 14



Weighting: 35% (35 Marks)

Important Note: Students who submit the assessment by the draft due date will receive up to 4 additional marks on top of the grades they score as bonus points.

Overview

This is a practical and real-world project that puts the knowledge you gained into practice. You are required to investigate and understand a publicly available dataset, design a conceptual model for storing the dataset in a relational database, apply normalisation techniques to improve the model, build the database according to your design and import the data into your database, and develop SQL queries in response to a set of requirements.

The objective of this assignment is to reinforce what you have learned in the whole course. Specifically, it involves how to build a simple application that connects to a database backend, running a simple relational schema.

Part A: Understanding the Data (0 Marks, Preliminary Work)

Part B: Designing the Database (10%)

- Task B.1 Produce an ER diagram for a relational database that will be able to store the given dataset.

Part C: Creating the Database and Importing Data (10%)

- Task C.1 Produce one SQL script file
- Task C.2 Create a database file and import the given dataset into your database.

Part D: Data Retrieval and Visualisation(15%)

- Task D.1-D.5 Produce one SQL query file that includes five SQL queries, produce a PDF file that includes the running result screenshot of five queries.
- Task D.1-D.5 Represent each query result as graph. Include graph in the PDF file for query result.
- Note that Task D.5 varies between ISYS1055 and ISYS3412. Only complete the version for your specific course.

Assessment criteria

This assessment will measure your ability to:

- Analyse the requirements outlined in the problem description
- Develop a conceptual model for the design of a database backend required for the system
- Use an industry-standard ER modelling tool to draw the ER model
- Use 7-step mapping process to create relational database schema
- Use normalisation process to evaluate the schema and make sure that all the relations are at least 3NF
- Create tables on SQLite Studio and populate them with data available from the specified sources.
- Write SQL statements required for CRUD (create, read, update and delete) operations on the database you built
- Develop your knowledge further to represent data in a meaningful way using data visualisation.

Course learning outcomes

This assessment is relevant to the following course learning outcomes:

CLO1	Describe the underlying theoretical basis of the relational database model and apply the theories into practice;
CLO2	Explain the main concepts for data modelling and characteristics of database systems.
CLO3	Develop a sound database design using conceptual modeling mechanisms such as entity-relationship diagrams.
CLO4	Develop a database based on a sound database design;
CLO5	Apply SQL as a programming language to define database schemas, update database contents, and to extract data from databases for specific users' information needs.
CLO6	Create then populate a normalised database based on a publicly available dataset, and visualise the results of queries against the dataset.

Part A: Understanding the Data

In this assignment, we are working with the publicly available dataset: **A Global Database of COVID-19 Vaccinations**. Further details about this dataset are available in the article available through the following URL: <https://www.nature.com/articles/s41562-021-01122-8>. The abstract of the article is as follows.

An effective rollout of vaccinations against COVID-19 offers the most promising prospect of bringing the pandemic to an end. We present the Our World in Data COVID-19 vaccination dataset, a global public dataset that tracks the scale and rate of the vaccine rollout across the world. This dataset is updated regularly and includes data on the total number of vaccinations administered, first and second doses administered, daily vaccination rates and population-adjusted coverage for all countries for which data are available (169 countries as of 7 April 2021). It will be maintained as the global vaccination campaign continues to progress. This resource aids policymakers and researchers in understanding the rate of current and potential vaccine rollout; the interactions with non-vaccination policy responses; the potential impact of vaccinations on pandemic outcomes such as transmission, morbidity and mortality; and global inequalities in vaccine access.

A live version of the vaccination dataset and documentation are available in a public GitHub repository at <https://github.com/owid/covid-19-data/tree/master/public/data/vaccinations>. These data can be downloaded in CSV and JSON formats.

For the purposes of completing this assignment, we are only using the following files. You are required to review and analyse the dataset available in these files. You will find that reviewing the rest of the files, even if not listed below, will help you to form a better understanding about the big picture.

FILE NAME	DESCRIPTION
1 locations.csv	Country names and the type of vaccines administered. Each line represents the last observation in a specific country. Refer to README.md for the details.
2 us_state_vaccinations.csv	History of observations for various locations in the US.
3 vaccinations-by-age-group.csv	History of observations for vaccinations of various age groups in each country.
4 vaccinations-by-manufacturer.csv	History of observations for various types of vaccines used in each country.
5 vaccinations.csv	Country-by-country data on global COVID-19 vaccinations. Each line represents an observation date. Refer to README.md for the details.
6 country_data/China.csv	Daily observations of vaccination in China.
7 country_data/India.csv	Daily observations of vaccination in India.
8 country_data/United States.csv	Daily observations of vaccination in US.
9 country_data/Ireland.csv	Daily observations of vaccination in Ireland.

Table 1: List of data files

To complete the tasks in the following sections, you are required to review and analyse the dataset that is available in the named files.

Part B: Designing the Database (10%)

Task B.1 Produce an ER diagram for a relational database that will be able to store the given dataset.

It is important to note that the given CSV files are not necessarily representing a good design for a relational database. It is your task to design a database that will adhere to good design principles that were taught throughout the course. This means your database schema will not match the structure of the CSV files and, therefore, you will require to manipulate the structure of the dataset (and not the data itself) to import it into your database. Importing the data is required to complete Task C.2.

The ER diagram must be produced by [Lucidchart](#) similar to the exercises that were completed in the course. UML notation is expected and using other notations will not be acceptable. Including a high-quality image representing your model is important, which can be achieved using Export function of Lucidchart.

You are also required to transform the ER diagram into a database schema that will be used in the next part of the assignment.

Creating a good database design typically involves some database normalisation activities. You should document your normalisation activities and support them with good reasoning. This typically involves explaining what the initial design was, what the problem was, and what changes have been made to rectify the issue.

The expected outcome of completing this task is one PDF file named Model.pdf containing the following sections.

1. Database ER diagram and, if needed, a reasonable set of assumptions.
2. Explanation of normalisation challenges and the resulting changes.
3. Database schema.

Part C: Creating the Database and Importing Data (10%)

Task C.1 Produce one SQL script file named Database.sql. This script file requires all the SQL statements necessary to create all the database relations and their corresponding integrity constraints as per your proposed design in Part B. The script file must run without any errors in SQLite Studio and contain necessary commenting to separate various relations. Note that this script is not supposed to store any data into the relations.

The expected outcome of completing this task is one script file with the specific name of Database.sql.

Task C.2 Create a database file named Vaccinations.db and import the given dataset into your database.

To complete this task, you may need to change the format of the CSV files to match the attributes of your designed database. You can use a spreadsheet editor such as Microsoft Excel.

The next step is to *import* the spreadsheets into the database you create in SQLite Studio. To complete this task, use the menu option *Tools – Import* in SQLite.

The expected outcome of completing this task is one database file named Vaccinations.db, which must contain all the data that is stored in the CSV files named in Table 1.

Part D: Data Retrieval and Visualisation (15%)

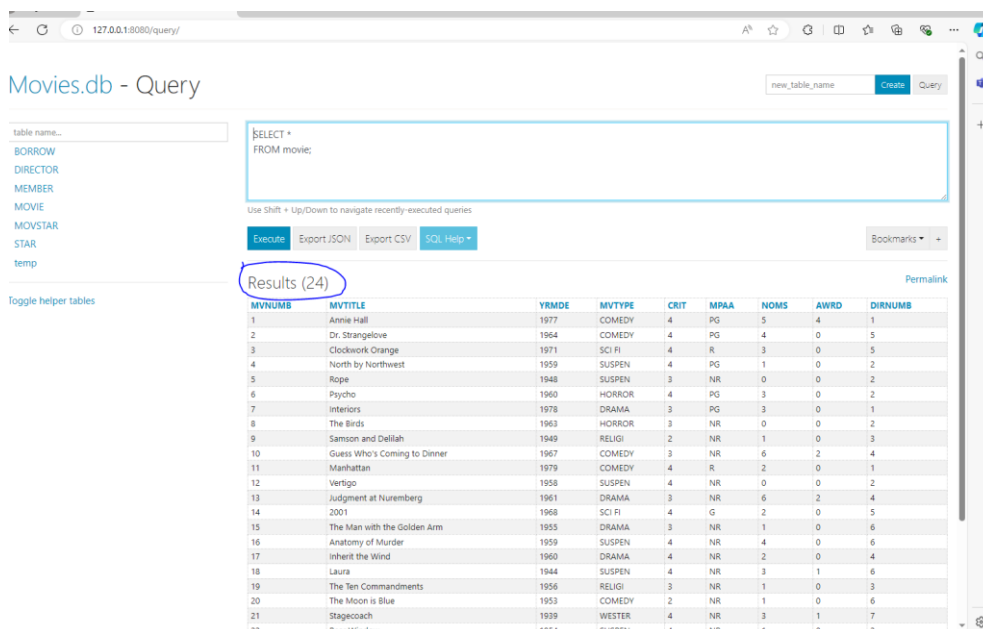
Now that you have created and populated a database, it is time to create some queries to investigate the data in various ways. In addition to writing the required queries, you are also asked to produce data visualisation for the results of your queries.

Each query must consist of one SQL statement. It would be acceptable to use several nested queries, combine several SELECT statements with various operators etc. However, it would not be acceptable to have multiple and separated queries for each task (or to use views).

After you have written each query, you are expected to inject your query into webpage, run the query on SQLite_Web, produce a data visualisation for each result set. You have the freedom to choose the tool for creating your visuals (e.g., Excel, Google Charts, Tableau) as well as the visualisation techniques (e.g., charts, plots, diagrams, maps). Completing this portion of the work will require that you understand the nature of the results of each query, undertake research to choose a visualisation tool you are comfortable with, decide about the best technique to visually represent each result set, and produce the visualisation. **Answers to tasks in Part D that are not supported by a visualisation can achieve up to 80% of the grade associated with each task.**

The expected outcome of completing this task is as follows.

1. One SQL script file named Queries.sql containing *all* the queries developed for the tasks in this section. It is important that you add comment lines to separate the queries and indicate which task they belong to. Note that valid SQL comments must not generate errors in SQLite Studio. The marker of your work will use this file to execute and test your queries.
1. A PDF file named Queries.pdf containing the following elements for each task.
 - a. The SQL query.
 - b. A snapshot of your query conducted on SQLite_Web. The snapshot must also show the total number of results retrieved by the query. A sample snapshot is provided below for your reference.



MVNUMB	MVTITLE	YRMD	MVTYPE	CRIT	MPAA	NOMS	AWRD	DIRNUMB
1	Annie Hall	1977	COMEDY	4	PG	5	4	1
2	Dr. Strangelove	1964	COMEDY	4	PG	4	0	5
3	Clockwork Orange	1971	SCI FI	4	R	3	0	5
4	North by Northwest	1959	SUSPEN	4	PG	1	0	2
5	Rope	1948	SUSPEN	3	NR	0	0	2
6	Psycho	1960	HORROR	4	PG	3	0	2
7	Interiors	1978	DRAMA	3	PG	3	0	1
8	The Birds	1963	HORROR	3	NR	0	0	2
9	Samson and Delilah	1949	RELIGI	2	NR	1	0	3
10	Guess Who's Coming to Dinner	1967	COMEDY	3	NR	6	2	4
11	Manhattan	1979	COMEDY	4	R	2	0	1
12	Vertigo	1958	SUSPEN	4	NR	0	0	2
13	Judgment at Nuremberg	1961	DRAMA	3	NR	6	2	4
14	2001	1968	SCI FI	4	G	2	0	5
15	The Man with the Golden Arm	1955	DRAMA	3	NR	1	0	6
16	Anatomy of Murder	1959	SUSPEN	4	NR	4	0	6
17	Inherit the Wind	1960	DRAMA	4	NR	2	0	4
18	Laura	1944	SUSPEN	4	NR	3	1	6
19	The Ten Commandments	1956	RELIGI	3	NR	1	0	3
20	The Moon is Blue	1953	COMEDY	2	NR	1	0	6
21	Stagecoach	1939	WESTER	4	NR	3	1	7
22	Rear Window	1954	SUSPEN	4	NR	1	0	2

Figure 1: Sample results snapshot with total rows

- c. Data visualisation. This must be represented as a graph or chart that presents the results in an easy to understand manner. Consider how to order or group the data to make it more meaningful visually.

List of Tasks

Task D.1 For any three given dates (i.e., you can assume any three dates, e.g., 1 Jan. 2021, 1 June 2021 and 1 Jan. 2022), list the dates, the total number of vaccines administered in each observation date in each of all countries, and the percentage change between the administered vaccines. The countries in resulting list should be ranked based on the total change in vaccine administration. Each row in the result set must have the following structure.

(Note: OD3 is after OD2, OD2 is after OD1). Column - total_vaccinations, 231 countries (3 marks)

Date 1 (OD1)	Country Name (CN)	Vaccine on OD1 (VOD1)	Date 2 (OD2)	Vaccine on OD2 (VOD2)	Date 3 (OD3)	Vaccine on OD3 (VOD3)	Percentage change of totals [(VOD2-VOD1)/VOD1- (VOD3-VOD2)/VOD2]
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Figure 2: Column Headers in the Result Set for Task D.1

Task D.2 Find the monthly growth rate of vaccine administered by each country, and compare each country's performance to the global average. Output growth rate of vaccine and the comparison for countries above the global average. Each row in the result set must have the following structure. (3 marks)

Country Name	Month	Year	Growth rate of vaccine (GR) $\left(\frac{\text{Cumulative Doses in this Month}}{\text{Cumulative Doses in previous Month}}\right)$	Difference of growth rate to global average (GR-average GR for all countries)
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Figure 3: Column Headers in the Result Set for Task D.2

Task D.3 Produce a list of the top 5 market share percentage of each vaccine type within each country, and the countries taking each of these vaccine types. For a vaccine type that has been taken in multiple countries, the result set is required to show several tuples reporting each country in a separate tuple. Each row in the result set must have the following structure. (3 marks)

Vaccine Type	Country	Percentage of vaccine type
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Figure 4: Column Headers in the Result Set for Task D.3

Task D.4 There are different data sources used to produce the dataset. Output the number of vaccines administered in each country for each month according to each data source (i.e., unique URL). Order the result set by the monthly administered vaccines. Each row in the result set must have the structure below. (3 marks)

join - locations summary details URL to daily or total vaccinations, order by

Country Name	Month	Source Name (URL)	Total Administered Vaccines
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Figure 5: Column Headers in the Result Set for Task D.4

Task D.5 How do various countries compare in the speed of their vaccine administration? Produce a report that lists all the observation days in 2022 and 2023, and then for each date, list the increment of total number of people fully vaccinated in each one of the 4 countries used in this assignment comparing with the previous date. (3 marks. this question is for ISYS1055 only; not required for ISYS3412)

Dates	United States	China	Ireland	India
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Figure 6: Column Headers in the Result Set for Task D.5

Task D.5 How does China track in terms of the speed of their vaccine administration? Produce a report that lists all the observation dates in 2022 and 2023, and then for each date, list the total number of people fully vaccinated in the top 3 countries which have the most fully vaccinated people in the observation dates from 2022 to 2023 taking each Vaccine type. (3 marks. this question is for ISYS3412 only; not required for ISYS1055)

Date	Vaccine Type	Fully Vaccinated people in Top 1 country	Fully Vaccinated people in Top 2 country	Fully Vaccinated people in Top 3 country
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Figure 7: Column Headers in the Result Set for Task D.5

Timeline:

Consider the following timeline to help manage your time to successfully complete this Project:

Timeline	Activities
Week 9 Part A	- Review this assignment, the assessment criteria & rubrics. - Familiarise yourself with the Harvard Referencing style. - Go to the website provided in the instructions for Part A to familiarise yourself with the dataset you are working with. - "A Global Database of COVID-19 Vaccinations". - Read the article for further details on this dataset. - Watch video/attend live session on how to get started, develop the ER model, create the relations, format the CSV file using Microsoft Excel and import the data into the database.
Week 10-11 Part B	- Work on Part B of the assessment. - Get up to date with Lucidchart to produce the ER diagram. - Create a PDF and create sections as mentioned in the assessment instructions for Part 2.
Week 12 Part C	- Work on Part C of the assessment + submit draft (part A + B + partial C). - Create relations, format CSV files according to database design and import data.
Week 13 Part D	- Work on Part D of the assessment. - Download a data visualisation tool that you will be using e.g. Excel, Google Charts, Tableau.
Week 14	Revise / fine-tune your assessment including the files required for submission.

Assessment Criteria

Your report will be assessed on the following criteria:

- This assessment will measure your ability to:
- Analyse the requirements outlined in the problem description
- Develop a conceptual model for the design of a database backend required for the system
- Use an industry-standard ER modelling tool to draw the ER model
- Use the 7-step mapping process to create relational database schema
- Use the normalisation process to evaluate the schema and make sure that all the relations are at least 3NF
- Create tables on SQLite Studio and populate them with data available from the specified sources
- Write SQL statements for CRUD (create, read, update, delete) operations on the database you built
- Develop your knowledge further to represent data in a meaningful way using data visualisation

[Learn how to use rubrics for assessment in Canvas.](#)

Submission Format

You are required to submit the files with the exact names as below.

1. Model.pdf
2. Database.sql
3. Vaccinations.db
4. Queries.sql
5. Queries.pdf

In the previous sections of the assignment, the expected content of each of the files is explained in detail.

Referencing guidelines

Use [RMIT Harvard](#) referencing style for this assessment.

You must acknowledge all the courses of information you have used in your assessments.

Refer to the [RMIT Easy Cite](#) referencing tools to see examples and tips on how to reference in the appropriated style. You can also refer to the library referencing page for more tools such as EndNote, referencing tutorials and referencing guides for printing.

Academic integrity and plagiarism

Academic integrity - When you submit work, it must be your own. Learn [how to avoid plagiarism and cheating](#)

Academic integrity is about honest presentation of your academic work. It means acknowledging the work of others while developing your own insights, knowledge, and ideas.

You should take extreme care that you have:

- Acknowledged words, data, diagrams, models, frameworks and/or ideas of others you have quoted (i.e., directly copied), summarised, paraphrased, discussed, or mentioned in your assessment through the appropriate referencing methods.
- Provided a reference list of the publication details so your reader can locate the source if necessary. This includes material taken from Internet sites.

If you do not acknowledge the sources of your material, you may be accused of plagiarism because you have passed off the work and ideas of another person without appropriate referencing, as if they were your own.

RMIT University treats plagiarism as a very serious offence constituting misconduct.

Plagiarism covers a variety of inappropriate behaviours, including:

- Failure to properly document a source
- Copyright material from the internet or databases
- Collusion between students

For further information on our policies and procedures, please refer to the [University website](#).

Assessment declaration

When you submit work electronically, you agree to the [assessment declaration](#).

Study support

Visit the 'Setting up for Success' module in the course to get help with referencing, writing skills, study skills, finding information, group work, and more. You can also get connected to one-on-one support with an Academic Skills Advisor, a librarian, or a peer mentor.